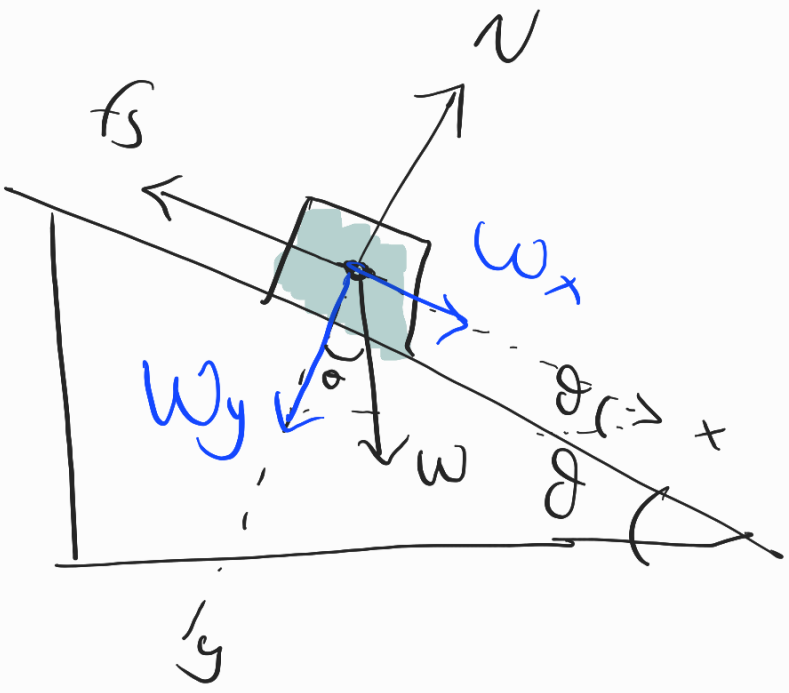
①



$\omega_x = \omega \sin\theta$

$\omega_y = \omega \cos\theta$

$\Sigma F_y = 0$

$\Rightarrow N = \omega_y$

$\Rightarrow N = \omega \cos\theta$

$N = mg\cos\theta_c$ ①

$\Sigma F_x = 0$

$\Rightarrow f_s = \omega_x$

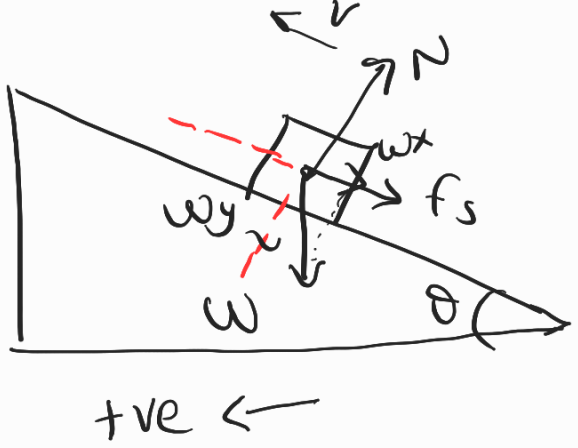
$\Rightarrow f_s = \omega \sin\theta = \omega \sin\theta_c$

$\Rightarrow \mu_s \cdot N = mg\sin\theta_c$ ②

① → ② : $mg\cos\theta_c \cdot \mu_s = mg\sin\theta_c$

$\Rightarrow \mu_s = \dfrac{\sin\theta_c}{\cos\theta_c} = \tan(\theta_c)$ //

②



$v_0 = 30 m/s$

$\theta = 15°$

$\mu_s = 0.7$

$v^2 = v_0^2 + 2ax$ ③

$\sum F_x = m \cdot \vec{a}$

$w_x = w \sin\theta$

$w_y = w \cos\theta$

$ma = -f_s - w_x$

$\Rightarrow m.a = -\mu_s . N - mg\sin\theta$ ①

$\sum F_y = 0 \Rightarrow N = w_y = mg\cos\theta$ ②

② → ①: $ma = -\mu_s \cdot mg\cos\theta - mg\sin\theta$

$\Rightarrow a = (-0.7)(10)\cos 15° - (10)\sin 15$

$= -7\cos 15 - 10\sin 15$

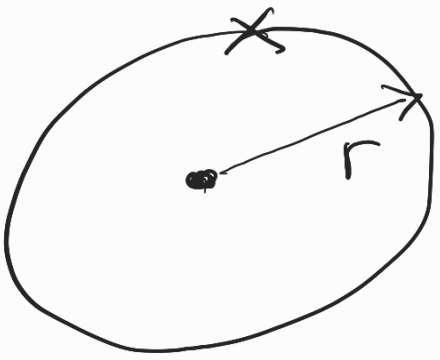
$= -7(0.97) - 10(0.26)$

$= -9.25 \, m/s^2$

$\Rightarrow$ (3) : $0^2 = 30^2 - 2(9.25)(x)$

$\Rightarrow x = \dfrac{30^2}{2(9.25)} = 48.6\text{m}$ //

③



$r = 6370 \times 10^3 \, m$

$T = ?$

$v = ?$

$a = 9.81 \, m/s^2$

$a = v^2/r \Rightarrow 9.81 = \frac{v^2}{6370 \times 10^3}$

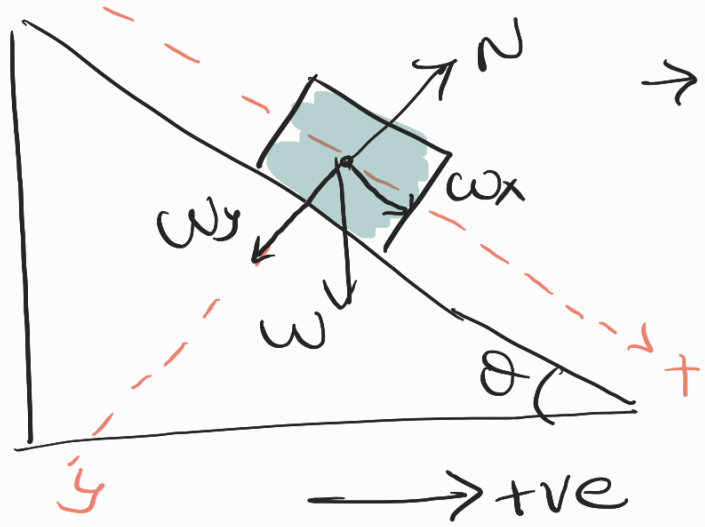
$\Rightarrow v = 7905 \, m/s$

$v = \frac{\Delta s}{\Delta t} \Rightarrow$ Εφόσον κάνει 1 περιστροφή τότε

$\Delta s = 2\pi r \equiv$ περίμετρος

$\Rightarrow v = \frac{2\pi r}{T} \Rightarrow T = \frac{2\pi r}{v} = 5063 \, s$ //

④



→ Όταν $\mu_s = 0$ ⇒ Μόνο βάρος και κάθετη αντίδραση

N ≡ κάθετη αντίδραση

w ≡ βάρος

$m = 950\,kg$

$v = 40\,km/h = 11.11\,m/s$

$r = 30\,m$

$w_y = w\cos\theta$

$w_x = w\sin\theta$

$\Rightarrow \sum F_x = \frac{mv^2}{r} = w_x$

$\Rightarrow w\sin\theta = mv^2/r$

$\Rightarrow mg\sin\theta = mv^2/r$

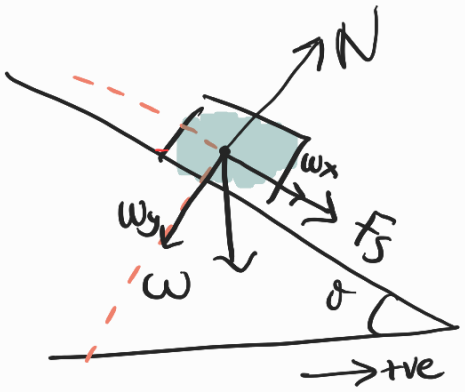
$\Rightarrow \sin\theta = v^2/gr$

$\Rightarrow \theta = \arcsin\left(\frac{v^2}{gr}\right)$

$\Rightarrow \theta = \arcsin\left(\frac{11.11^2}{10 \cdot 30}\right)$

$\Rightarrow \theta \simeq 23^\circ$ //

$\mu_s = 0.3$



→ για μέγιστη ταχύτητα η τριβή είναι ίση το μέγεθος

→ $F_s \equiv$ στατική τριβή

$\Rightarrow W_y = W\cos\theta$

$W_x = W\sin\theta$

$\Sigma F_x = mv^2/r = W_x + F_s$

$\Rightarrow mg\sin\theta + \mu_s N = mv^2/r$ ①

$\Sigma F_y = 0 \Rightarrow N = W_y$

$\Rightarrow N = W\cos\theta = mg\cos\theta$ ②

② → ①: $mg\sin\theta + \mu_s mg\cos\theta = mv^2/r$

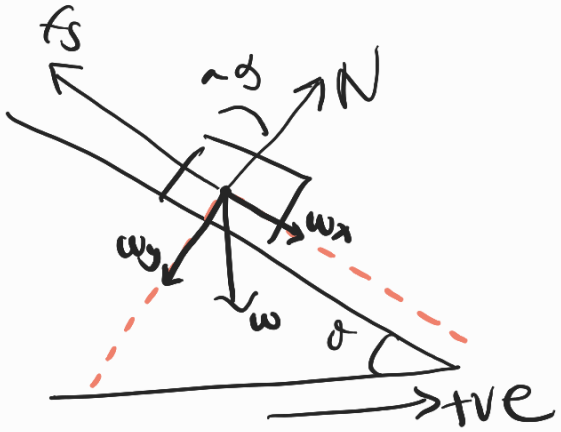
$\Rightarrow gr(\sin\theta + \mu_s\cos\theta) = v^2$

$\Rightarrow v = \sqrt{gr(\sin\theta + \mu_s\cos\theta}$

$\Rightarrow v = \sqrt{10\,30(\sin 23 + 0.3\cos 23)}$

$\Rightarrow v = 15.9\,m/s$ → μέγιστη ταχύτητα

→ για ελάχιστη ταχύτητα η τριβή είναι αντίθετη στην επιφάνεια.



$$\sum F_x = \omega_x - f_s = mv^2/r$$

$$\Rightarrow mg\sin\theta - \mu_s \cdot N = mv^2/r$$

$$\Rightarrow mg\sin\theta - \mu_s \cdot mg\cos\theta = mv^2/r$$

↳ από (2)

$$\Rightarrow gr(\sin\theta - \mu_s\cos\theta) = v^2$$

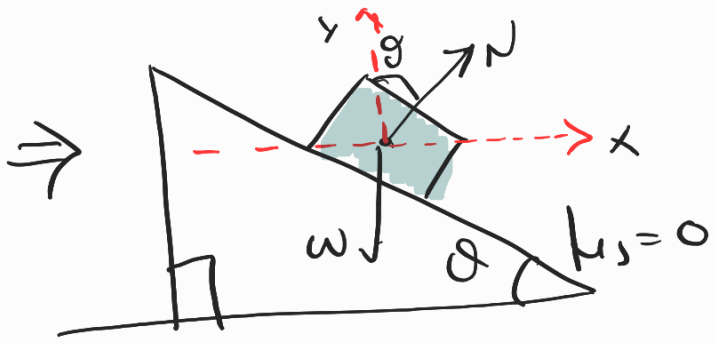
$$\Rightarrow v = \sqrt{gr(\sin\theta - \mu_s\cos\theta)}$$

$$\Rightarrow v = \sqrt{10 \cdot 30(\sin 23 - 0.3\cos 23)} \simeq 6.05\ m/s$$

↓

ελάχιστη ταχύτητα

④ ΕΝ ΑΠΑΧΤΙΚΗ ΠΥΖΗ



$$\Rightarrow N_y = N \cdot \cos\vartheta$$

$$N_x = N \cdot \sin\vartheta$$

$$\Sigma F_x = \frac{mv^2}{r} \Rightarrow N_x = mv^2/r$$

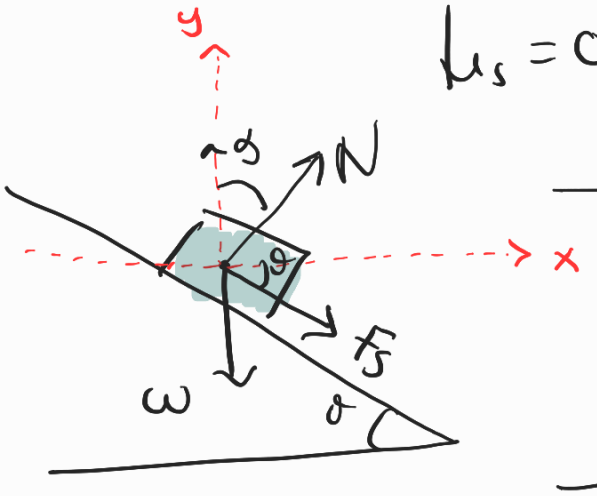
$$\Rightarrow N\sin\vartheta = mv^2/r \quad (1)$$

$$\Sigma F_y = 0 \Rightarrow N_y = W$$

$$\Rightarrow N\cos\vartheta = m \cdot g \quad (2)$$

$$\frac{(1)}{(2)} : \frac{N\sin\vartheta}{N\cos\vartheta} = \frac{\frac{mv^2}{r}}{mg} \Rightarrow \tan\vartheta = \frac{v^2}{rg}$$

$$\Rightarrow \vartheta = \arctan\left(\frac{11.11^2}{30 \cdot 10}\right) \simeq 23^\circ \;//$$



$\mu_s = 0.3$

→ για μέγιστη ταχύτητα η τριβή είναι ίσως το μέγιστο

→ $F_s \equiv$ στατική τριβή

$F_{sx} = F_s \cos\vartheta$

$F_{sy} = F_s \sin\vartheta$

$\sum F_x = mv^2/r \Rightarrow F_{sx} + N_x = mv^2/r$

$\Rightarrow F_s \cos\vartheta + N \sin\vartheta = mu^2/r$

$\Rightarrow \mu_s N \cos\vartheta + N \sin\vartheta = mu^2/r$

$\Rightarrow N(0.3\cos\vartheta + \sin\vartheta) = mu^2/r$ ③

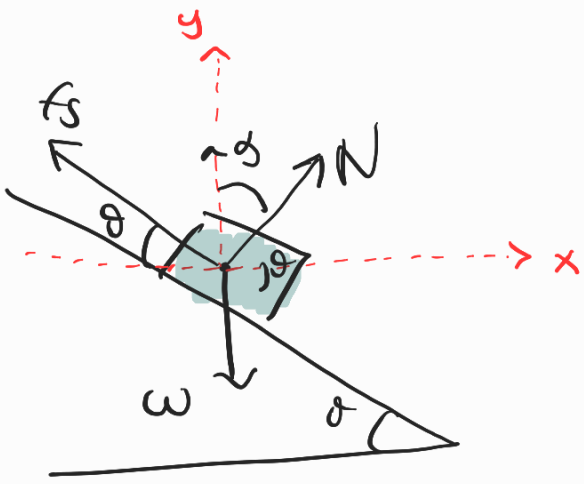
$\sum F_y = 0 \Rightarrow w + F_{sy} = N_y$

$\Rightarrow -F_s \sin\vartheta + N\cos\vartheta = +mg$

$\Rightarrow N(\cos\vartheta - 0.3\sin\vartheta) = mg$ ④

$$\frac{③}{④}: \quad \frac{N(0.3\cos\vartheta + \sin\vartheta)}{N(\cos\vartheta - 0.3\sin\vartheta)} = \frac{\frac{mu^2}{r}}{\frac{mg}{1}}$$

$$\Rightarrow v_{max} = \sqrt{\frac{rg(0.3\cos\vartheta + \sin\vartheta)}{\cos\vartheta - 0.3\sin\vartheta}} \approx 15.9 \text{ m/s} \quad //$$



→ για ελάχιστη ταχύτητα η τριβή είναι αριθμητικά στο περιθώριο.

$$\Rightarrow \Sigma F_x = N_x - f_{sx} = mv^2/r$$

$$\Rightarrow N\sin\vartheta - f_s\cos\vartheta = mv^2/r$$

$$\Rightarrow N(\sin\vartheta - 0.3\cos\vartheta) = mv^2/r \quad (5)$$

$$\Sigma F_y = N_y + f_{sy} - w = 0$$

$$\Rightarrow N\cos\vartheta + f_s\sin\vartheta = mg$$

$$\Rightarrow N(\cos\vartheta + 0.3\sin\vartheta) = mg \quad (6)$$

$$\frac{(5)}{(6)}: \frac{N(\sin\vartheta - 0.3\cos\vartheta)}{N(\cos\vartheta + 0.3\sin\vartheta)} = \frac{\frac{mv^2}{r}}{\frac{mg}{1}}$$

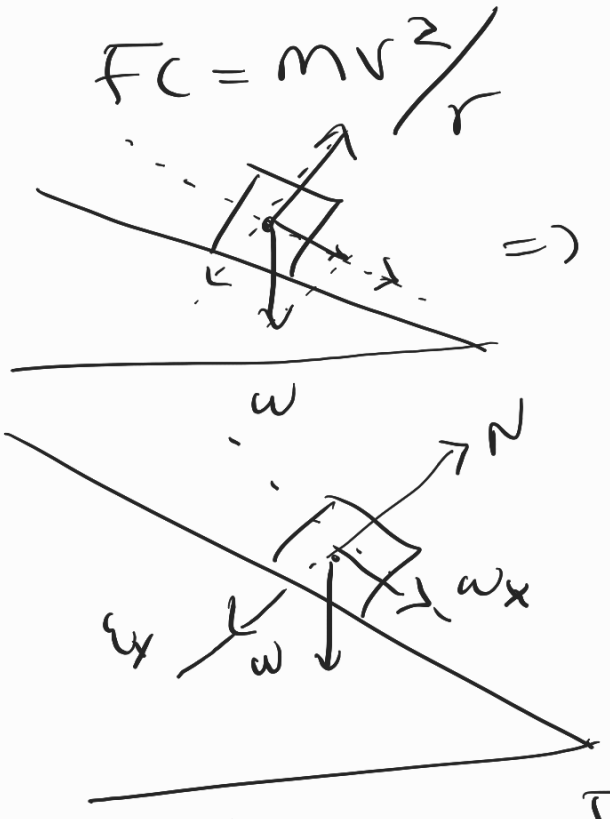
$$\Rightarrow v_{min} = \sqrt{\frac{Rg(\sin\vartheta - 0.3\cos\vartheta)}{\cos\vartheta + 0.3\sin\vartheta}}$$

$$\simeq 6.05 \text{ m/s} \;//$$

if V slow.

$\Rightarrow F_c$ required is less

$F_c = mv^2/r$



$\Rightarrow$ If slow $F_c <<$

$\Rightarrow W_x$ pulls in centre.

because $w_x > F_c$

$\Rightarrow F_s$ pulls outwards

⑤ $t_u = 0.95\ s \rightarrow f = 4.5\ Hz$

$t_d = 0.24\ s \rightarrow v = 0\ m/s$

(a) $a_{up} = ?$ $a_{down} = ?$

spin-up: $\omega_0 = 0$

$\omega_f = 2\pi f = 2\pi(4.5)$

$$a_{up} = \frac{d\omega}{dt} = \frac{\omega_f - \omega_0}{t_u} = \frac{2\pi(4.5)}{0.95}$$

$$\Rightarrow a_{up} = 29.76\ rad/s^2$$

spin-down: $\omega_0 = 2\pi f$

$\omega_f = 0$

$$\Rightarrow a = \frac{0 - 2\pi f}{t_d} = \frac{-2\pi(4.5)}{0.24}$$

$$= -117.81\ rad/s^2 \ //$$

(b) $\omega_{max} = ?$

$$\omega = 2\pi f = 2\pi(4.5) = 9\pi\ rad/s$$

$$= 28.27\ rad/s \ //$$

(c) $x = u_0 t + \frac{1}{2} a t^2$

$\Rightarrow \boxed{\vartheta = \omega_0 t + \frac{1}{2} a t^2}$ → γωνία ως διάνυσμα

spin-up: $\vartheta = 0 + \frac{1}{2}(0.95)^2(29.76)$

$\Rightarrow \vartheta \simeq 13.42 \text{ rad}$

$\Rightarrow n = \frac{13.42}{2\pi} \simeq 2.14$ περιστροφές

spin-down:

$\vartheta = 2\pi f (0.24) + \frac{1}{2}(-117.8)(0.24)^2$

$= (28.27)(0.24) - \frac{1}{2}(117.8)(0.24)^2$

$= 3.39 \text{ rad}$

$\Rightarrow n = \frac{3.39}{2\pi} \simeq 0.54$ περιστροφές

$\Rightarrow r = \frac{2.14}{0.54} = 3.96$ //